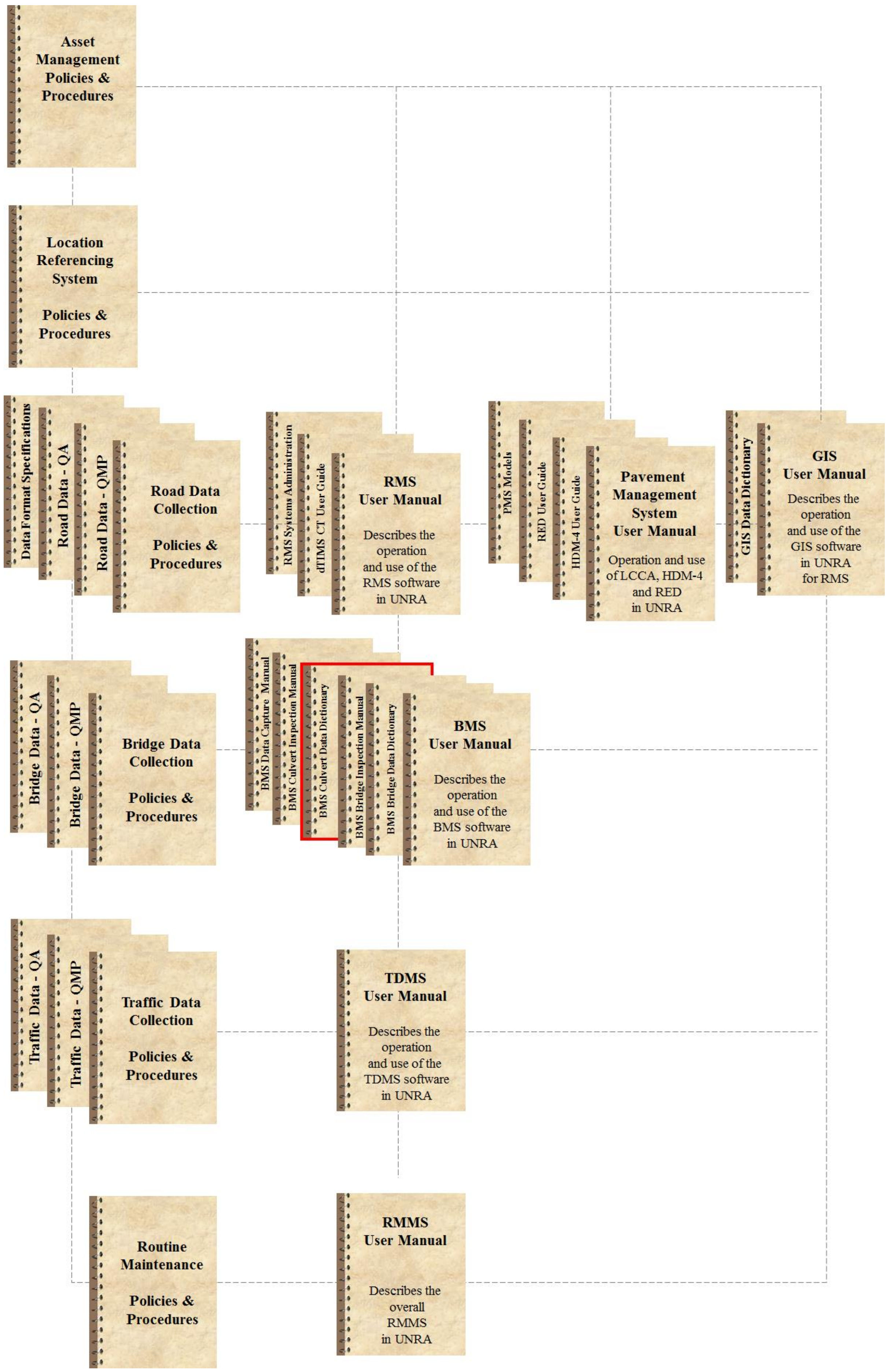
# Uganda National Roads Authority

## Asset Management System

Culvert Management System (CMS) Data Dictionary

***Asset Management System Documentation Map***



Asset Management Policies & Procedures
Location Referencing System
Policies & Procedures
Data Format Specifications
Road Data - QA
Road Data - QMP
Road Data Collection
Policies & Procedures
RMS Systems Administration
dTIMS CT User Guide
RMS User Manual
Describes the operation and use of the RMS software in UNRA
PMS Models
RED User Guide
HDM-4 User Guide
Pavement Management System User Manual
Operation and use of LCCA, HDM-4 and RED in UNRA
GIS Data Dictionary
GIS User Manual
Describes the operation and use of the GIS software in UNRA for RMS
Bridge Data - QA
Bridge Data - QMP
Bridge Data Collection
Policies & Procedures
BMS Data Capture Manual
BMS Culvert Inspection Manual
BMS Culvert Data Dictionary
BMS Bridge Inspection Manual
BMS Bridge Data Dictionary
BMS User Manual
Describes the operation and use of the BMS software in UNRA
Traffic Data - QA
Traffic Data - QMP
Traffic Data Collection
Policies & Procedures
TDMS User Manual
Describes the operation and use of the TDMS software in UNRA
Routine Maintenance
Policies & Procedures
RMMS User Manual
Describes the overall RMMS in UNRA

This document contains the BMS Data Dictionary for the Uganda National Roads Authority (UNRA). It forms part of the Asset Management System documentation as shown on the previous page.

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# TABLE OF CONTENTS

# 1. DEFINITION OF INVENTORY DATA FIELDS

## 1.1 CULVERT NUMBER (#)

**DESCRIPTION:** This is a unique number assigned to each culvert.

**FORMAT :** Character String : Maximum length of 6 characters.

This is a compulsory field. It is very important to use the correct culvert number. This culvert number is the identification field in the Culvert Management System (CMS). Prefix "TC" will be culverts which do not have an official registered culvert number (#) yet.

Examples:

| | | | | | |
|---|---|---|---|---|---|
| C | 0 | 2 | 1 | 2 | |
| C | 0 | 5 | 6 | 2 | |
| C | 3 | 7 | 9 | 0 | A |
| C | 0 | 0 | 7 | 8 | |
| C | 5 | 1 | 2 | 5 | |

## 1.2 LINK NAME OR LINK NUMBER

**DESCRIPTION:** This field indicates the link identification of the link name of the road crossing over the culvert.

**FORMAT :** Character String : Maximum length of 255 characters.

Example: Abim – Kilak, Aduku – Lira, Agwata - Kachu, etc.

*Note: 1) Leave this field blank if the feature crossing over the culvert is not a road.*

## 1.3 KM DISTANCE

**DESCRIPTION:** This field indicates the km position of the culvert.

**FORMAT :** Decimal String : Range between 0,00 and 999,99.

Km distances must always refer to the permanent RMS km reading. This could differ from the km distances used during the construction phase. For this reason, the km distance should be verified during inspections.

*Note: If the principal feature is not a road, this item must be left blank.*
*If the km distance is unknown, it must be left blank.*

## 1.4 CULVERT TYPE

**DESCRIPTION:** This field indicates the type of culvert.

**FORMAT :** Integer String : The culvert type is described by a two digit code

**01** Box culvert (in situ concrete)
**02** Box culvert (precast units)
**03** Concrete pipe culvert
**04** Corrugated metal pipe culvert
**05** Corrugated metal arch (Armco) culvert
**06** Concrete arch culvert
**98** Unknown

## 1.5 NUMBER OF PIPES / CELLS

**DESCRIPTION:** This field indicates the number of spans / cells or openings.

**FORMAT** : Integer Field : Range between 1 and 99

This is a compulsory field.

## 1.6 SPAN / DIAMETER

**DESCRIPTION:** This field indicates the span length or diameter of a specific culvert cell / opening

**FORMAT** : Decimal String : Range between 0,0 and 99,99

For cell structures, the span length is measured perpendicular to the centre line of the culvert and should only be the clear width per opening. In case of more than one type of opening, the largest opening should be recorded and mention should be made of the actual culvert size in the remarks table.

## 1.7 HEIGHT

**DESCRIPTION:** This field indicates the height of wall or rise of opening

**FORMAT** : Decimal String : Range between 0,0 and 99,9

The height of wall is measured from the invert level to the soffit level, on the inlet side of the culvert. If the height varies significantly along the culvert length, details thereof can be supplied in the remarks table.

## 1.8 GIS CO-ORDINATES (DECIMAL DEGREES)

**DESCRIPTION:** This field indicates the respective E and S co-ordinates of the centre point of the culvert in decimal degrees.

**FORMAT :** Decimal Field : Range between 0.00000 and 99.99999.

## 1.9 MAINTENANCE STATION

**DESCRIPTION:** This field gives the Maintenance Station of the culvert.

**FORMAT :** String : The Maintenance Station is a string:

**?** Unknown
**01** Kampala
**02** Mpigi
**03** Masaka
**04** Mubende
**05** Luwero
**06** Mbarara
**07** Kabale
**08** Kasese
**09** Fort Portal
**10** Hoima
**11** Masindi
**12** Lira
**13** Gulu
**14** Kitgum
**15** Moyo
**16** Arua
**17** Jinja
**18** Tororo
**19** Mbale
**20** Soroti
**21** Moroto
**22** Kotido

## 1.10 INVENTORY PHOTOS

A minimum of 9 inventory photos shall be taken, ie

- Photo 1: General elevation photo from upstream
- Photo 2: View from approach along centreline of deck, in direction of increasing kilometres

- Photo 3: Photo of culvert number plate, or culvert number as provided on culvert, or alternatively of area on culvert where number is typically provided (even if no number is provided)
- Photo 4: Underside of deck
- Photo 5: Typical wing wall
- Photo 6: Typical parapet / guardrail
- Photo 7: Feature crossed

Inventory photos shall have the following photo file names (YY refers to year, eg 15):

- Photo 1: C............._YY_01.jpg
- Photo 2: C............._YY_02.jpg
- Photo 3: C............._YY_03.jpg
- Photo 4: C............._YY_04.jpg
- Photo 5: C............._YY_05.jpg
- Photo 6: C............._YY_06.jpg
- Photo 7: C............._YY_07.jpg

## 1.11 DESIGN APPROVAL DATE

**DESCRIPTION:** Use this field to indicate the year when the design drawings were approved by the responsible authority.

*Note: The Design Approval Date is to be given in the format: YYYY, and can be found on the design drawings or "As Built" drawings.*

## 1.12 DESIGN CODE

**DESCRIPTION:** Use this field to indicate the Code of Procedure that was used in the design of the culvert.

**FORMAT :** Integer String : The Design Code is indicated by a two-digit code:

**01** SATCC (1988)
**02** BS5400
**03** CEB/FIP
**04** DIN
**05** CP114
**06** TMH7 (1988)
**07** BS153
**08** Uganda Bridge Code
**98** Other
**?** Unknown

## 1.13 COMPLETION YEAR

**DESCRIPTION:** Use this field to indicate the year when construction was completed. This date can be obtained from the culvert number (#) plates during inspection or from as-built drawings.

*Note: Use the "As Built" date on the drawings when the culvert number (#) plates are not available. The completion date is to be given in the format : YYYY.*

## 1.14 DATE MAJOR REHABILITATION

**DESCRIPTION:** This field indicates the completion date of major rehabilitation work performed on the culvert.

**FORMAT :** Integer String : YYYY

The date of major rehabilitation works is to be indicated in the format YYYY. Rehabilitation is defined as work aimed at upgrading the general condition of a culvert or the load carrying capacity of a culvert (e.g. widening). Please give a description of the rehabilitation in the block for additional remarks.

## 1.15 DESIGN LIVE LOADING

**DESCRIPTION:** This field indicates the characteristic live load, as defined in various loading codes, for which the culvert was designed.

**FORMAT :** Integer String : The type of design load is described by a two digit code.

**01** Full M.O.T./B.M.T. - Not tested for NC
**02** Full M.O.T./B.M.T. - NC30
**03** Full M.O.T./B.M.T. - Less than NC30
**04** NA + NB24
**05** NA + NB36 (NC not tested)
**06** NA + NB36 + NC
**07** HA + HB25
**08** HA + HB36
**09** HA + HB45
**10** Pedestrian only
**11** RU (rail)
**12** Uganda Bridge Code
**98** Other
**?** Unknown

## 1.16 DESIGN RETURN PERIOD (YEARS)

**DESCRIPTION:** This field indicates the return period of the design flood.

**FORMAT :** Integer String : Range between 0 and 999.

## 1.17 DESIGN FREEBOARD (M)

**DESCRIPTION:** This field indicates the freeboard corresponding to the design flood for the culvert, measured from the shoulder break point of the road.

**FORMAT :** Decimal String : Range between 0,0 and 99,9.

## 1.18 DESIGN FLOOD DISCHARGE ($m^3/s$)

**DESCRIPTION:** This field indicates the magnitude of the design flood for the culvert.

**FORMAT :** Integer String : Range between 0 and 99999.

$1m^3/s = 1ft^3/s \times 0.3048^3$

## 1.19 CULVERT GRADIENT (%)

**DESCRIPTION:** This field indicates the longitudinal gradient along the culvert invert.

**FORMAT :** Decimal String : Range between 0 and 89,9.

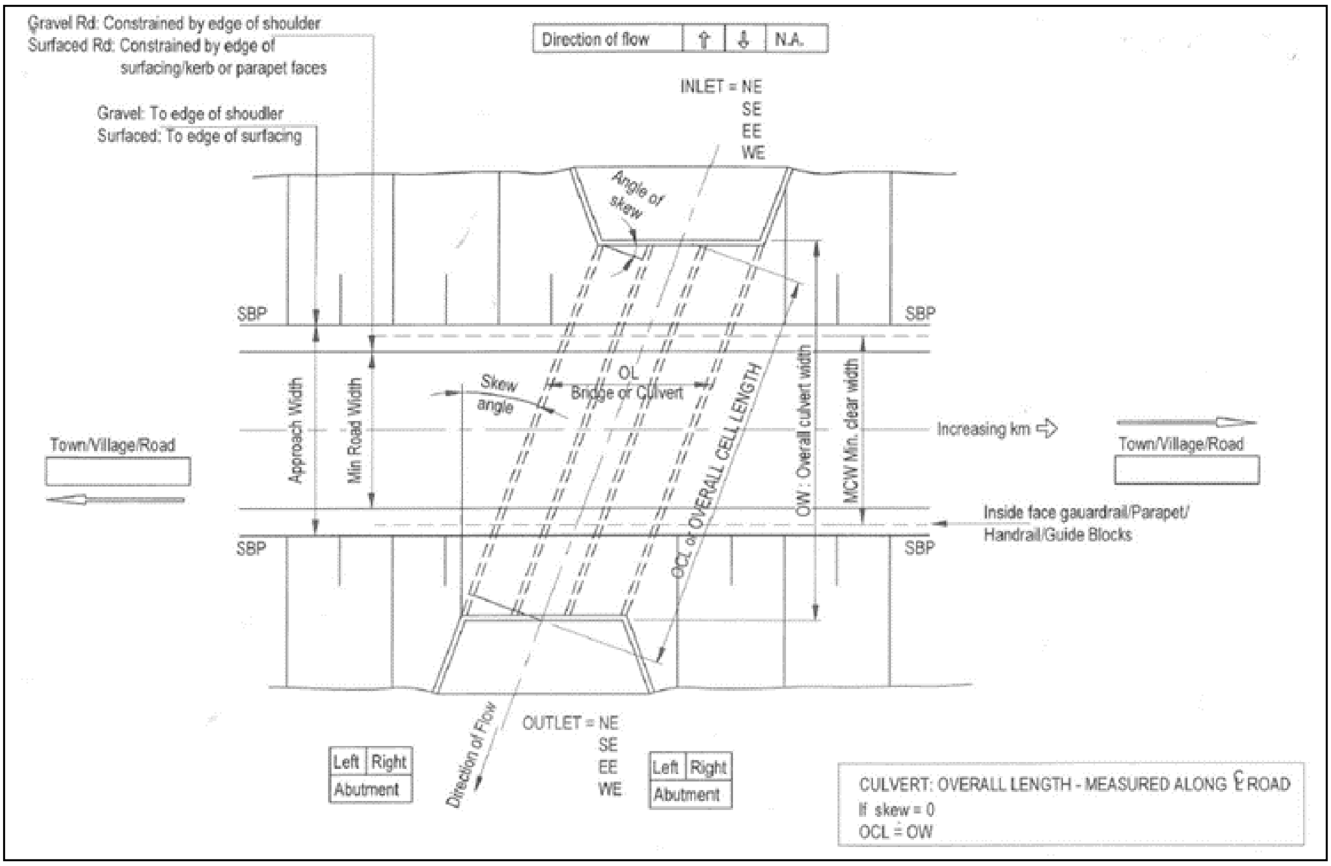


Figure 1: Dimensions of a culvert

## 1.20 OVERALL LENGTH (m) ('OL' IN FIG 1)

**DESCRIPTION:** This field indicates the overall length of the culvert.

**FORMAT :** Decimal String : Range between 0,0 and 999,9.

For box/pipe/cell culverts the overall length is measured parallel to the centreline of the road to the edges of the outer walls.

## 1.21 OVERALL WIDTH (m) ('OW' IN FIG 1)

**DESCRIPTION:** This field indicates the overall width of the culvert perpendicular to the road.

**FORMAT :** Decimal String : Range between 0,0 and 999,9.

For box/pipe/cell culverts the overall width is measured perpendicular to the road.

## 1.22 MINIMUM CLEAR WIDTH (m) ('MCW' IN FIG 1)

**DESCRIPTION:** This field indicates the minimum clear width on the road surface between barriers or kerbs.

**FORMAT :** Decimal String : Range between 0,0 and 999,9.

For box/pipe/cell culverts the minimum clear width is measured perpendicular to the road normally between barriers or kerbs. If there are no barriers or kerbs use the value equal to Minimum road width.

## 1.23 OVERALL CELL LENGTH (m) ('OCL' IN FIG 1)

**DESCRIPTION:** This field indicates the overall cell length of the culvert along the centre line of the culvert.

**FORMAT :** Decimal String : Range between 0,0 and 999,9.

For box/pipe/cell culverts the overall cell length is the length of the culvert measured in the direction of the centre line of the culvert. It is the length measured from inlet to outlet.

## 1.24 MINIMUM ROAD WIDTH (m) (REFER FIG 1)

**DESCRIPTION:** This field indicates the minimum road width perpendicular to the road.

**FORMAT :** Decimal String : Range between 0,0 and 999,9.

For box/pipe/cell culverts the minimum road width is measured perpendicular to the centre line of the road as follows:

Surfaced roads: As constrained by edge of surfacing, kerbs, inner edges of guardrails or parapet faces.
Gravel roads: As constrained by edge of shoulder.

## 1.25 APPROACH WIDTH (m) (REFER FIG 1)

**DESCRIPTION:** This field indicates the approach width.

**FORMAT :** Decimal String : Range between 0,0 and 999,9.

For box/pipe/cell culverts the approach road width is measured perpendicular to the centre line of the road as follows:

Surfaced roads: As constrained between shoulder break points (same as formation width).

Gravel roads: As constrained between shoulder break points (same as formation width).

## 1.26 SKEW ANGLE (DEGREES) (REFER FIG 1)

**DESCRIPTION:** This field indicates the skew angle of the culvert with respect to a line perpendicular to the road direction.

**FORMAT :** Decimal String : Range between 0,0 and 99,9.

For a structure that is 90° to the centerline of the road the angle of skew is 0°.

## 1.27 MAX FILL HEIGHT (m)

**DESCRIPTION:** This field indicates the maximum height of road fill material on top of a culvert.

**FORMAT :** Decimal String : Range between 0,0 and 99,9.

The height of fill refers to the maximum fill height above the culvert along the culvert length.

## 1.28 MIN VERTICAL CLEARANCE (m)

**DESCRIPTION:** This is the vertical opening remaining between the top of debris and the top slab of the culvert.

**FORMAT :** Decimal String : Range between 0,0 and 99,9.

This is the vertical opening remaining between the top of debris and the top slab of the culvert. If there is no debris inside the culvert it is the same as the “MAXIMUM CELL SIZE - HEIGHT (m)”.

## 1.29 MAXIMUM CELL SIZE - WIDTH (m)

**DESCRIPTION:** This is the width of a single cell measured perpendicular to the walls of the culvert.

**FORMAT :** Decimal String : Range between Range between 0,0 and 99,9

This is the width of a single cell measured perpendicular to the walls of the culvert.

## 1.30 MAXIMUM CELL SIZE - HEIGHT (m)

**DESCRIPTION:** This is the height measured between the constructed invert of the culvert and the top slab of the culvert.

**FORMAT :** Decimal String : Range between 0,0 and 99,9.

This is the height measured between the constructed invert of the culvert and the top slab of the culvert. Generally, if there is no debris inside the culvert the “MAXIMUM CELL SIZE - HEIGHT (m)” and the “MIN VERTICAL CLEARANCE (M)” are the same.

## 1.31 MIN. DEPTH OF FILL OVER (m).

**DESCRIPTION:** This is minimum height of fill measured above the structure
.

**FORMAT :** Decimal String : Range between 0,0 and 99,9.

This is minimum height of fill measured above the structure.